

Talagrand's inequality does not force Poincaré after the quadratic-tail map

Automated mathematics discovery run `fa_banach_001`
Lane 2, candidate for human review

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Status: candidate counterexample, likely valid. We construct an absolutely continuous probability measure μ on $\mathbb{R}$ which satisfies Talagrand's quadratic transport inequality $\mathbf{T}_2$, while $(\omega_2)_\# \mu$ does not satisfy Poincaré inequality. This answers the necessity question after Corollary 9.15 of arXiv:1003.3852 negatively.

1 Source question

Gozlan and Léonard define

$$\omega_2(x) = \operatorname{sgn}(x) \max\{|x|, |x|^2\}$$

and prove that Poincaré inequality for $(\omega_2)_\# \mu$ implies $\mathbf{T}_2$ for μ . Immediately afterward they ask whether the condition is necessary [1, Corollary 9.15 and the following paragraph, PDF p. 54].

We skip the technical proof of this inequality and refer to [51, Lemma 2.6 and Proof of Proposition 4.2]. Let us emphasize an important particular case. In the sequel, $\omega_2 : \mathbb{R} \rightarrow \mathbb{R}$ will be the function defined by $\omega_2(x) = \max(x, x^2)$, for all $x \geq 0$ and such that $\omega_2(-x) = -\omega_2(x)$, for all $x \in \mathbb{R}$.

Corollary 9.15. *If a probability measure μ on $\mathbb{R}^k$ verifies the inequality $\mathbf{P}(\omega_2, C)$ for some $C > 0$ then it verifies the inequality $\mathbf{T}_2(4\omega_2(\tau\sqrt{kC}))$, where τ is some universal constant.*

In other words, a sufficient condition for μ to verify $\mathbf{T}_2$ is that the image of μ under the map ω_2 verifies Poincaré inequality. We do not know if this condition is also necessary.

Source PDF page 54; the definition, corollary, and exact question are visible.

2 Definitions and counterexample

Write

$$d\lambda(t) = \frac{1}{2}e^{-|t|} dt, \quad a_n = 16^n \quad (n \geq 1),$$

and, for $t \geq 0$, define the unit ramp

$$r_n(t) = \min\{\max\{t - a_n, 0\}, 1\}.$$

Set

$$b(t) = \sqrt{t+1} - 1, \quad F(t) = b(t) + \sum_{n \geq 1} r_n(t),$$

where the sum is locally finite. Extend F oddly to

$$T(t) = \operatorname{sgn}(t)F(|t|),$$

and let

$$\mu = T_{\#}\lambda.$$

Theorem 1 (Sparse-ramp counterexample). *The probability measure μ is symmetric, absolutely continuous, and has a positive density. It satisfies Talagrand's quadratic transport inequality T_2 . Nevertheless $(\omega_2)_{\#}\mu$ does not satisfy Poincaré inequality.*

3 Proof intuition

The two-sided exponential law has a transport inequality with the mixed cost $\min\{|s - t|^2, |s - t|\}$. A map which is both Lipschitz and Hölder of order $1/2$ turns this mixed cost into a quadratic cost, hence produces a T_2 measure. The square-root background $b(t)$ has exactly these two regularity properties.

The sparse unit ramps do not destroy either property: exponential spacing means that an interval of length d meets at most $O(\sqrt{d})$ ramps. But a ramp far out in the tail makes the reciprocal-density integral in the weighted Muckenhoupt criterion large. Its exponential factor cancels the tail probability, leaving a factor comparable to a_n , which diverges.

4 Regularity of the transport map

Lemma 1. *The odd homeomorphism T is $3/2$ -Lipschitz and satisfies*

$$|T(t) - T(s)| \leq 2\sqrt{2}|t - s|^{1/2} \quad (s, t \in \mathbb{R}).$$

Proof. The ramps have disjoint interiors. Almost everywhere on $[0, \infty)$,

$$0 < b'(t) \leq \frac{1}{2}, \quad 0 \leq \sum_n r'_n(t) \leq 1.$$

Thus F is $3/2$ -Lipschitz. Since $F(0) = 0$, the odd extension has the same Lipschitz constant, including across zero.

For the half-Hölder bound, take $0 \leq s < t$ and put $d = t - s$. The background satisfies $b(t) - b(s) \leq \sqrt{d}$. If $d < 1$, the interval $[s, t]$ meets at most one ramp interval and the total ramp increment is at most $d \leq \sqrt{d}$. Suppose $d \geq 1$ and N ramp intervals meet $[s, t]$. If $N = 1$, their total increment is at most $1 \leq \sqrt{d}$. If $N \geq 2$ and the first such interval has index p , then

$$d \geq a_{p+N-1} - (a_p + 1) \geq 16(16^{N-1} - 1) - 1 \geq N^2.$$

The ramp increment is at most $N \leq \sqrt{d}$. Hence $F(t) - F(s) \leq 2\sqrt{d}$ on either half-line. For opposite signs,

$$|T(t) - T(s)| \leq 2(\sqrt{|t|} + \sqrt{|s|}) \leq 2\sqrt{2}|t - s|^{1/2}.$$

Strict monotonicity, continuity, and divergence follow from b , so T is a homeomorphism. $\square$

5 Proof of Theorem 1

The quadratic transport inequality

The source recalls that λ satisfies a transport–entropy inequality for

$$c_0(s, t) = \min\{|s - t|^2, |s - t|\}$$

with a finite universal constant [1, Section 9.3]. Put $\delta = |s - t|$ and $\Delta = |T(s) - T(t)|$. Lemma 1 gives

$$\delta^2 \geq \frac{4}{9}\Delta^2, \quad \delta \geq \frac{1}{8}\Delta^2.$$

Consequently

$$c_0(s, t) \geq \frac{1}{8}|T(s) - T(t)|^2.$$

The contraction principle for transport inequalities therefore shows that $\mu = T_{\#}\lambda$ satisfies a quadratic transport–entropy inequality, i.e. T_2 (with an inessential finite change of constant).

Since T is locally absolutely continuous and $T'(t) > 0$ almost everywhere, μ has a positive density h . On the positive half-line it obeys

$$h(T(t)) = \frac{\frac{1}{2}e^{-t}}{T'(t)} \quad \text{for almost every } t > 0. \quad (1)$$

Symmetry follows from oddness of T .

Failure of Poincaré after ω_2

The median of μ is zero. Proposition 9.16 of the source says that $(\omega_2)_{\#}\mu$ satisfies Poincaré inequality if and only if, in particular,

$$D_{\omega_2}^+ = \sup_{x \geq 0} \mu[x, \infty) \int_0^x \frac{\omega_2'(u)^2}{h(u)} du < \infty. \quad (2)$$

Let $x_n = T(a_n + 1)$ and abbreviate $\beta_n = b(a_n) = \sqrt{a_n + 1} - 1$. Monotonicity gives

$$\mu[x_n, \infty) = \frac{1}{2}e^{-(a_n+1)}.$$

On the ramp $a_n < t < a_n + 1$, one has $T'(t) \geq 1$ and $T(t) \geq \beta_n > 1$. Hence $\omega_2'(T(t)) = 2T(t)$ there. Using (1) and the change of variables $u = T(t)$,

$$\begin{aligned} \int_0^{x_n} \frac{\omega_2'(u)^2}{h(u)} du &\geq \int_{a_n}^{a_n+1} 8T(t)^2 T'(t)^2 e^t dt \\ &\geq 8\beta_n^2 e^{a_n} (e - 1). \end{aligned}$$

Multiplying the last two displays yields

$$\mu[x_n, \infty) \int_0^{x_n} \frac{\omega_2'(u)^2}{h(u)} du \geq 4(1 - e^{-1})\beta_n^2 \longrightarrow \infty.$$

Thus $D_{\omega_2}^+ = \infty$, so $(\omega_2)_{\#}\mu$ fails Poincaré inequality. This proves the theorem. $\square$

6 Verification, limitations, and novelty check

The accompanying script `code/verify_sparse_ramp.py` samples 100,000 pairs around both signs and the first five ramps. It checks the Lipschitz, half-Hölder, and mixed-cost lower bounds and prints the first five Muckenhoupt lower bounds. The proof above is symbolic and does not depend on the computation.

This is a one-dimensional negative answer to the literal necessity question. No quantitative claim is made about the best T_2 constant. The map is piecewise smooth rather than globally smooth; the source question only asks about probability measures, and the resulting density is positive and absolutely continuous.

The run registry and bounded primary-source searches used arXiv:1003.3852, the exact source sentence, the authors, and combinations of “ ω_2 image”, “Talagrand T_2 ”, “weighted Poincaré”, “necessary”, and “counterexample”. They found the source survey and general weighted-Poincaré papers but no explicit answer or this sparse-ramp construction. This is not an exhaustive novelty certification.

Human review recommendation. High-priority specialist review, focusing on Lemma 1, the transport-cost comparison, and the Muckenhoupt change of variables.

References

- [1] N. Gozlan and C. Léonard, *Transport Inequalities. A Survey*, Markov Processes and Related Fields 16 (2010), 635–736; arXiv:1003.3852.